

A2. Community Detection Analysis in Synthetic and Real-World Networks

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Complex Networks

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1 Introduction

This report explores how communities are organized in complex networks by comparing the performance of three algorithms: Infomap, Louvain, and the Greedy algorithm. In the first part, we use synthetic networks generated with the Stochastic Block Model (SBM) and test how accurately these algorithms recover a known "ground truth" as the internal link density (p_{rr}) increases. To measure their performance, we use the Jaccard Index, Normalized Mutual Information (NMI), and Variation of Information (VI). In the second part, we analyze a real-world network of face-to-face interactions in a French primary school to see how contact duration (edge weights) affects the detected communities. By comparing these results with school metadata and discussing the theoretical limits of modularity, we highlight the strengths and weaknesses of these detection tools in both controlled and empirical settings.

2 Methodology

This section describes the computational framework and metrics used to analyze the community structure of both synthetic and real-world networks. By applying different algorithmic approaches, we aim to evaluate their consistency and accuracy across varying topological conditions.

2.1 Data Sources and Preparation

We analyzed two distinct datasets:

- **Synthetic Networks (SBM):** A series of networks generated using the Stochastic Block Model (SBM) with $N = 300$ nodes divided into five equal blocks. The inter-block probability was fixed at $p_{rs} = 0.02$, while the intra-block probability (p_{rr}) varied from 0.00 to 1.00 to simulate different levels of community cohesion.
- **Real-World Network (Primary School):** A network of face-to-face interactions in a French primary school. We analyzed both the *weighted* version (using cumulative interaction time) and the *unweighted* version (topology only) to understand the impact of edge intensity on social grouping.

2.2 Community Detection Algorithms

Three main algorithmic strategies were implemented to identify the mesoscale structure of the networks:

- **Modularity Maximization:** We used the **Louvain**, **Leiden**, and **Greedy Modularity** (Clauset-Newman-Moore) algorithms. These methods seek to maximize the modularity index (Q), which measures the density of links inside communities compared to a random null model.
- **Information-Theoretic:** The **Infomap** algorithm was used, which identifies communities by compressing the description of information flow (random walks) across the network.

2.3 Evaluation and Comparison Metrics

To evaluate the quality of the detected partitions, especially when a "ground truth" was available (Task 1), we employed the following metrics:

- **Accuracy Metrics:** Normalized Mutual Information (NMI), Jaccard Index, and Normalized Variation of Information (NVI) were used to compare detected communities against the true block structure.
- **Structural Metrics:** The modularity value (Q) and the total number of detected communities were recorded for every iteration to track the algorithms' stability.

3 Characterization of Community Structure in Networks with Block Structure

3.1 Performance Evolution vs. Intra-block Probability (p_{rr})

This section analyzes the structural transition of the synthetic networks as the internal block density (p_{rr}) increases. The results illustrate a clear shift from a random regime to a well-defined community structure, which is captured differently by each algorithm.

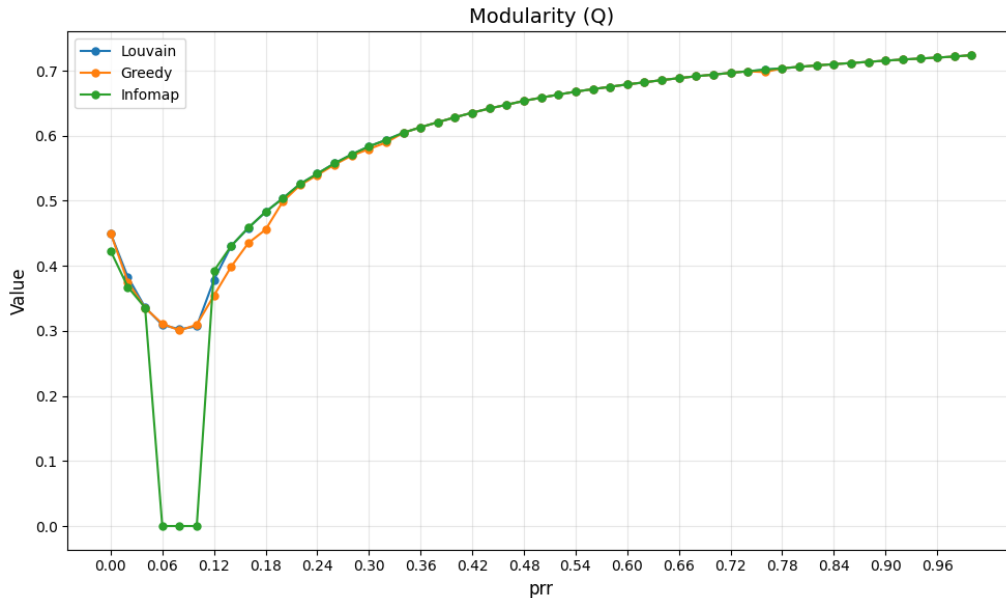


Figure 1: Evolution of the Modularity index (Q) for Louvain, Greedy, and Infomap as a function of p_{rr} .

As shown in Figure 1, the modularity (Q) generally increases for all three algorithms as p_{rr} grows. However, at low values, the algorithms overestimate the community structure; for example, at $p_{rr} = 0.02$, the modularity starts at approximately $Q = 0.38$. Modularity begins

to increase consistently once prr reaches 0.12. For all three algorithms, the modularity reaches its maximum value when $prr = 1.00$.

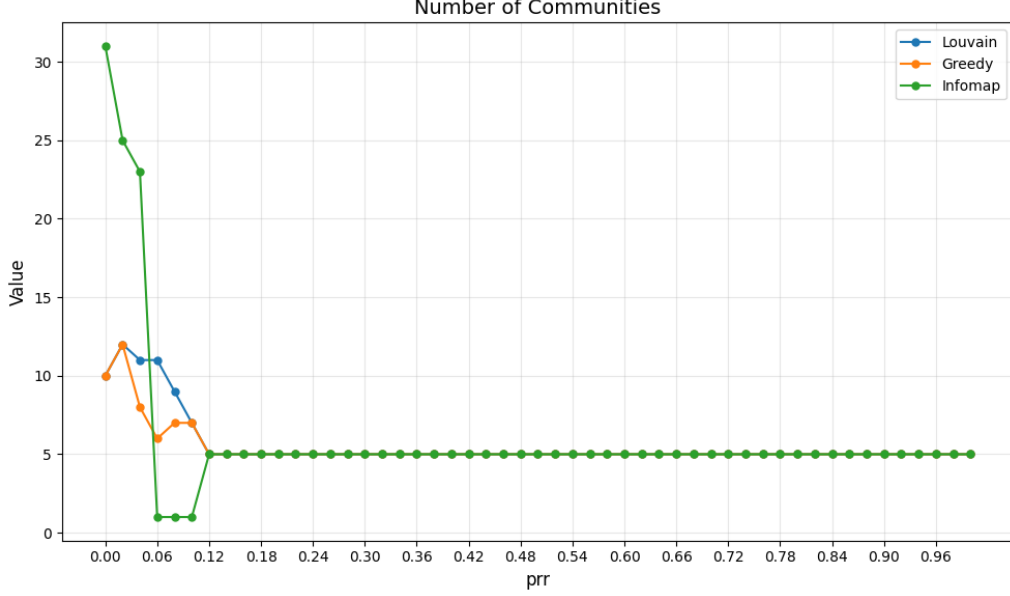


Figure 2: Number of communities detected by each algorithm.

The transition is further clarified by observing the number of detected communities in Figure 2. In the low-density regime ($prr < 0.10$), all algorithms exhibit significant instability. They tend to "over-partition" the network, detecting up to more than 30 communities. In this stage, the algorithms interpret random local fluctuations as community boundaries because the global block structure is too weak to be detected.

A sharp transition occurs between $prr = 0.10$ and 0.12 . In this range, the number of communities drops abruptly and stabilizes at exactly **5**, matching the ground truth.

3.2 Validation against Ground Truth

To evaluate the accuracy of the identified partitions, the detected communities were compared against the ground truth using Normalized Mutual Information (NMI), the Jaccard Index, and Normalized Variation of Information (NVI).

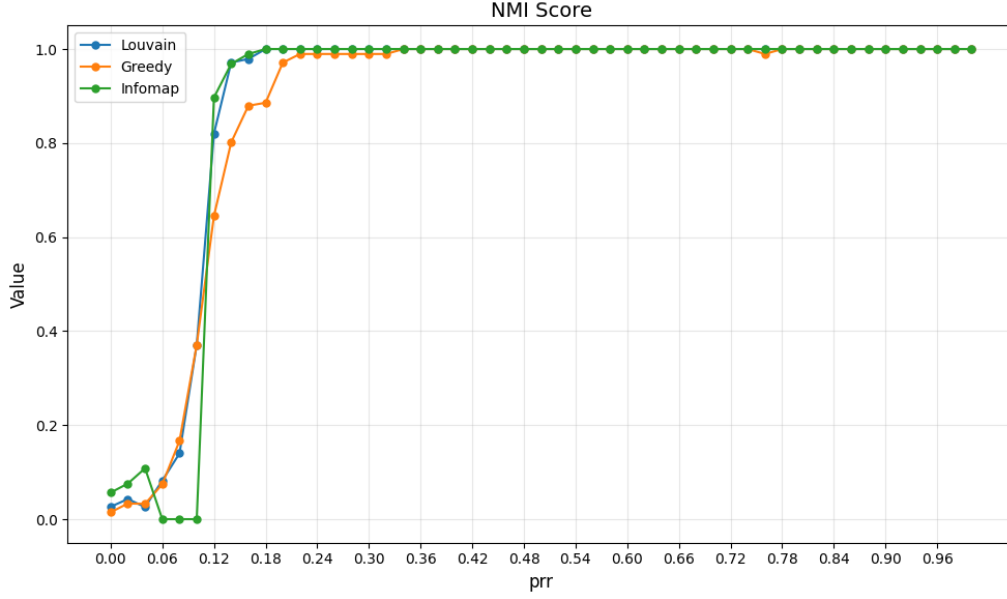


Figure 3: Normalized Mutual Information (NMI) vs. prr .

As shown in Figure 3, the NMI score starts at low values for $prr = 0.02$. However, consistent with the modularity transition, a sharp increase occurs around $prr = 0.12$. At this point, the NMI for all three algorithms rises to 1.0, indicating a perfect match with the ground truth blocks. This perfect similarity is maintained for all higher values of prr up to 1.00.

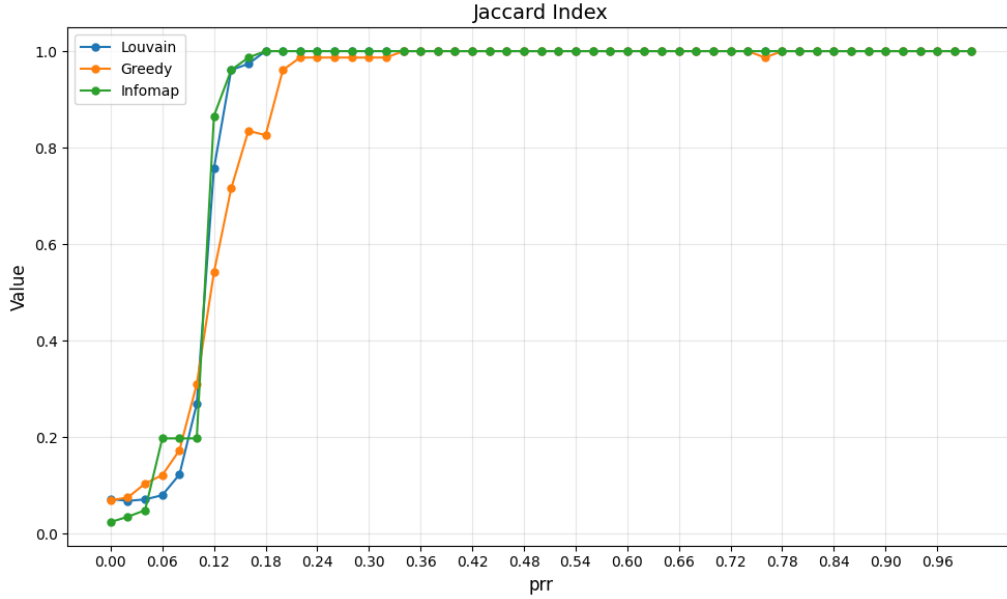


Figure 4: Jaccard Index vs. prr .

As shown in Figure 4, the Jaccard Index follows a trend similar to the other metrics, jumping abruptly at $prr = 0.12$. From this threshold onward, all algorithms reach a Jaccard Index of 1.0, confirming that the node-pair assignments in the detected communities are identical to the ground truth.

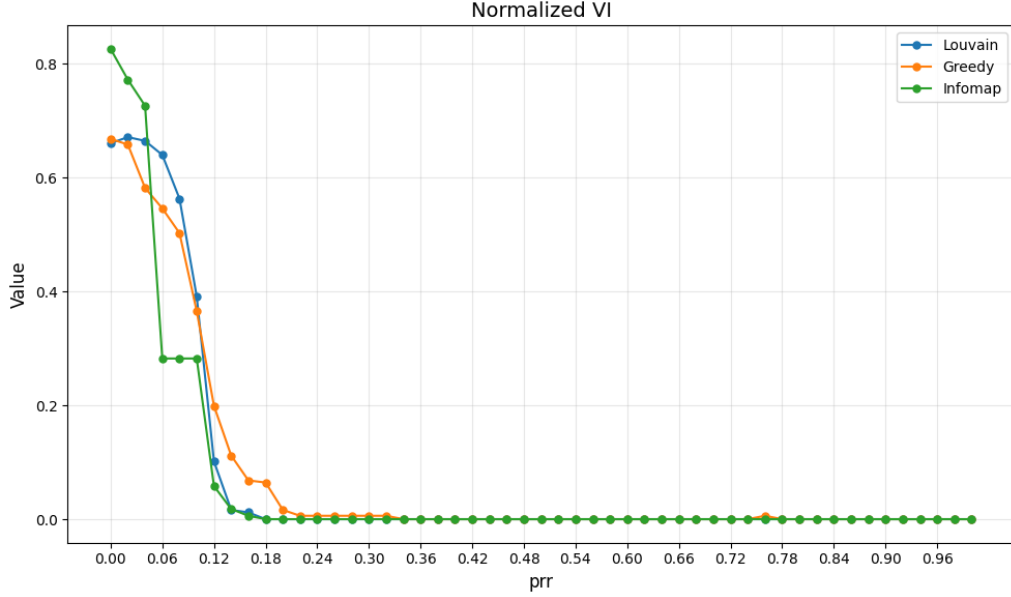


Figure 5: Normalized Variation of Information (NVI) vs. prr .

Finally, Figure 5 presents the Normalized Variation of Information (NVI). Because this metric measures the distance between partitions, high values represent low similarity. At $prr = 0.02$, the NVI is near 0.8 for all algorithms. Following the same transition observed in the previous metrics, the NVI drops sharply, reaching 0.0 at $prr = 0.18$. This indicates that the information lost or gained when comparing the detected partition to the ground truth becomes zero, signifying total convergence.

3.3 Visual Representation of Community Structure

This section provides a visual analysis of the network’s structural transition across three critical stages: the initial random state ($prr = 0.02$), the transition point where communities are first identified ($prr = 0.16$), and the final fully connected state ($prr = 1.0$). To ensure consistency, a fixed layout generated from the densest network ($prr = 1.0$) was used for all visualizations.

3.3.1 Initial State: $prr = 0.02$

In the initial state, the network exhibits a lack of clear community structure as the intra-block probability is equal to the inter-block noise. As shown in Figure 6, the nodes appear as a dense, uniform cluster where the ground-truth blocks cannot be visually distinguished.

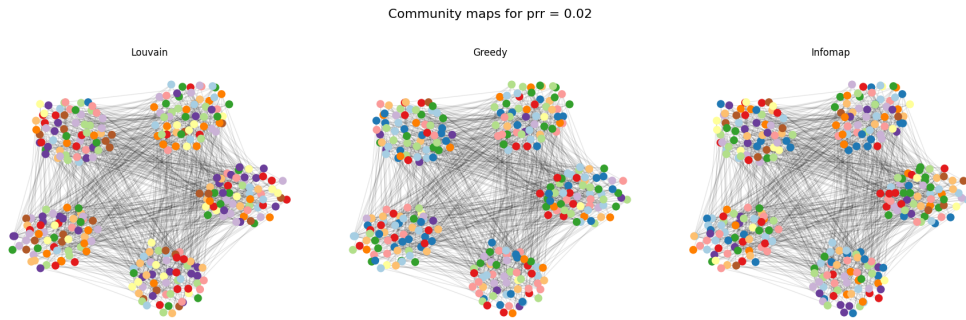


Figure 6: Community detection snapshots for $prr = 0.02$.

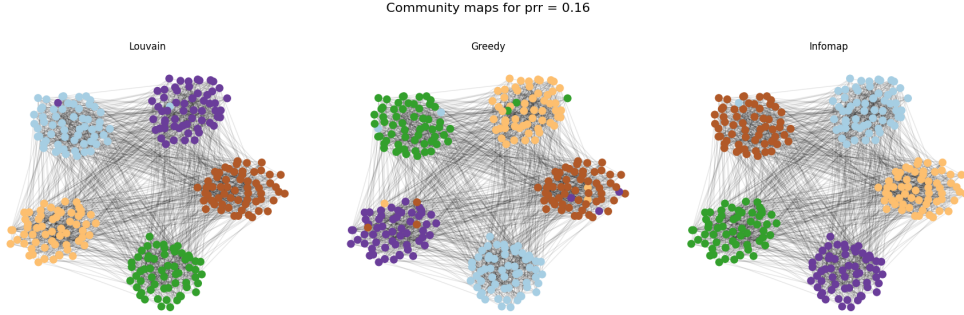
Table 1: Metrics for $prr = 0.02$

Algorithm	NMI	Modularity (Q)	Jaccard	NVI	Communities
Louvain	0.0429	0.3828	0.0680	0.6709	12
Greedy	0.0334	0.3753	0.0749	0.6578	12
Infomap	0.0758	0.3675	0.0351	0.7718	25

The metrics in Table 1 reflect this lack of structure. The algorithms overestimate the number of communities, and the low NMI and Jaccard values confirm that the detected partitions do not match the ground truth. Notably, even though the modularity values reach nearly 0.4, the network lacks a meaningful community structure, a finding that highlights the limitations of modularity as an absolute measure of partitioning quality.

3.3.2 Transition State: $prr = 0.16$

At $prr = 0.16$, the community structure becomes clearly defined. As seen in Figure 7, the algorithms successfully group the nodes into five distinct colored clusters that correspond to the ground-truth blocks.

Figure 7: Community detection snapshots for $prr = 0.16$.Table 2: Metrics for $prr = 0.16$

Algorithm	NMI	Modularity (Q)	Jaccard	NVI	Communities
Louvain	0.9789	0.4584	0.9737	0.0119	5
Greedy	0.8794	0.4352	0.8347	0.0681	5
Infomap	0.9894	0.4587	0.9866	0.0060	5

According to Table 2, all algorithms correctly identify exactly five communities. The similarity metrics jump toward 1.0, with Louvain and Infomap showing near-perfect recovery of the original block structure.

3.3.3 Dense State: $prr = 1.0$

Finally, at $prr = 1.0$, the internal links are saturated. The five communities are perfectly isolated and visually distinct in Figure 8, showing absolute convergence.

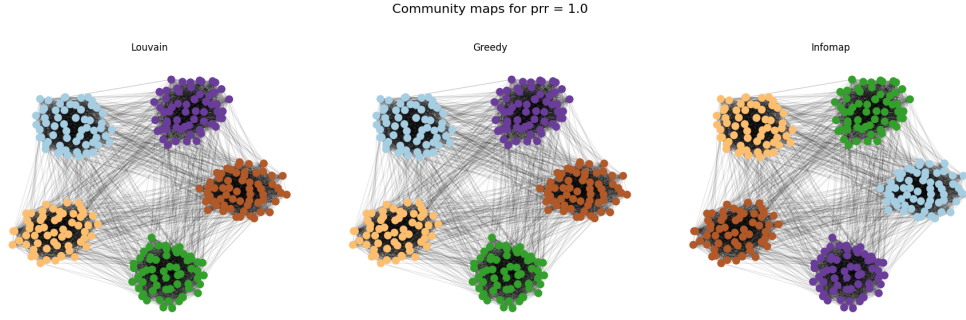


Figure 8: Community detection snapshots for $prr = 1.0$.

Table 3: Metrics for $prr = 1.0$

Algorithm	NMI	Modularity (Q)	Jaccard	NVI	Communities
Louvain	1.0	0.724	1.0	0.0	5
Greedy	1.0	0.724	1.0	0.0	5
Infomap	1.0	0.724	1.0	0.0	5

Table 3 confirms that in this state, all algorithms achieve perfect accuracy ($NMI = 1.0$, $Jaccard = 1.0$, $NVI = 0.0$), perfectly matching the ground truth.

3.4 Discussion: Limitations of Modularity

A central question in this analysis is whether a modularity value of $Q = 0.4$ is enough to confirm the existence of community structure. Based on our results, the answer is **no**. Modularity is a measure of relative edge density, but it does not inherently guarantee that the identified groups are meaningful or accurate.

Our findings from **3.3 Visual Representation of Community Structure**, specifically the data in Table 1, provide clear evidence of this limitation. At $prr = 0.02$, where the network is essentially random, the Louvain and Greedy algorithms reported modularity values near 0.38. To an observer looking only at Q , this would suggest a moderate community structure. However, the external metrics (NMI and Jaccard) contradict this, remaining near zero. This occurs because modularity maximization algorithms can exploit random fluctuations in edge distribution to create partitions even when no real structure exists.

In our evaluation, modularity only begins to align with the true ground-truth partitions after the transition point of $prr = 0.12$. This demonstrates that in sparse or weakly structured networks, Q can be misleadingly high due to the resolution limit or random noise. Therefore, modularity should not be used as a standalone proof of community structure; it must always be interpreted alongside external validation metrics.

3.5 Discussion: Algorithmic Differences

The comparison between Infomap, Louvain, and Greedy reveals that while all algorithms eventually converge to the ground truth, they respond differently to the presence of noise in the network.

- **Infomap:** This algorithm shows the highest sensitivity to local structure. As seen in Table 1, it detects 25 communities in the random regime, which is significantly more than the other methods. While this sensitivity leads to over-partitioning in noisy settings, it also allows Infomap to reach the highest accuracy (NMI of 0.9894) as soon as the community signal emerges at $prr = 0.16$.

- **Louvain:** This method provides a very balanced performance. It is less prone to extreme over-partitioning than Infomap in noisy conditions and reaches a near-perfect alignment with the ground truth (NMI of 0.9789) very quickly once prr exceeds 0.12. Its ability to consistently maximize modularity makes it a reliable and stable choice across most ranges of prr .
- **Greedy Modularity:** This algorithm is the most conservative of the three. It is slower to recognize the underlying block structure, as evidenced by its lower NMI (0.8794) and Jaccard index at the transition state of $prr = 0.16$. Although it is less "sharp" during the transition phase, it eventually reaches total convergence at high prr values.

In summary, the transition point where the algorithms begin to accurately detect the 5-block structure occurs between $prr = 0.12$ and 0.16 . This threshold marks where the signal-to-noise ratio becomes sufficient for the algorithms to overcome random fluctuations and identify the true mesoscale of the network. Each algorithm’s performance reflects its specific sensitivity: Infomap prioritizes flow and local detail, while Louvain and Greedy focus on global density thresholds.

4 Characterization of Community Structure in Real-World Networks

4.1 Comparative Analysis: Unweighted vs. Weighted Networks

This section compares the community structures detected in the binary unweighted network (G_u) and the weighted network (G_w). In the weighted version, edge weights represent the total duration of face-to-face interactions, while in the unweighted version, any interaction—regardless of duration—is treated as a simple link.

Table 4: Comparison of Modularity (Q) and Number of Communities (N_c) for Unweighted and Weighted Networks.

Algorithm	Unweighted (G_u)		Weighted (G_w)	
	N_c	Modularity (Q)	N_c	Modularity (Q)
Greedy	3	0.2284	6	0.6742
Louvain	6	0.2840	6	0.6727
Leiden	6	0.2840	6	0.6755

As shown in Table 4, incorporating interaction duration leads to a significant increase in modularity across all algorithms. While the unweighted modularity values remain below 0.3, the weighted modularity scores exceed 0.67. This suggests that interaction time is a much stronger indicator of social organization than the mere existence of contact. In a school environment, unweighted edges often represent "noisy" or accidental interactions, such as students passing in a hallway. By adding weights, the algorithms prioritize long-term interactions, effectively filtering out this noise and revealing a much more robust community structure.

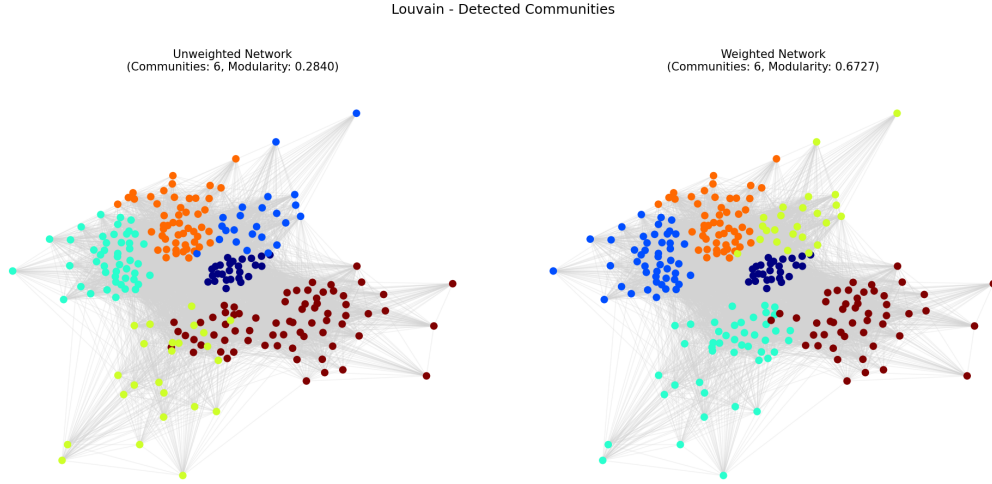


Figure 9: Community comparison for the Louvain algorithm.

The visual results for the **Louvain** algorithm (Figure 9) and the **Leiden** algorithm (Figure 11) show that while they detect 6 communities in both cases, the clusters in the weighted network are significantly more cohesive. In the unweighted plots, community boundaries are blurred, and several nodes are assigned to groups that are spatially distant in the layout. In contrast, the weighted partitions form tight, well-separated clusters.

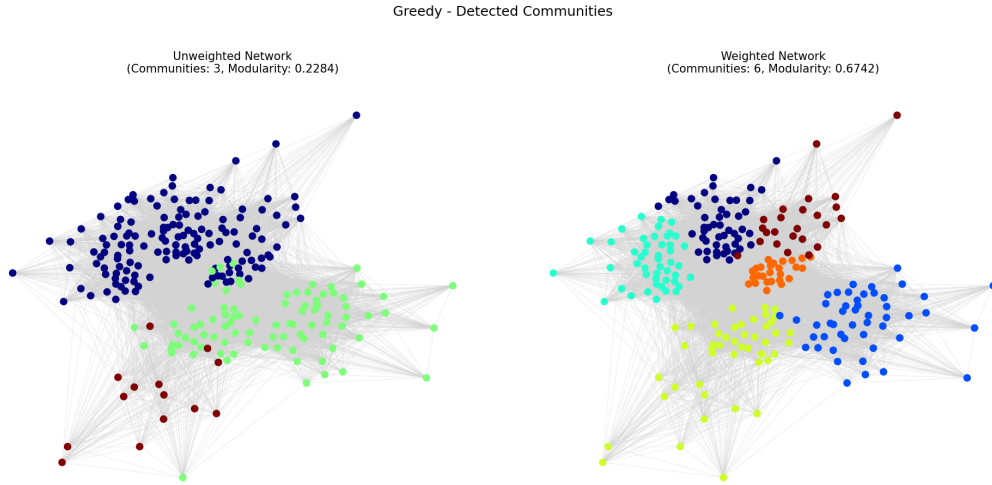


Figure 10: Community comparison for the Greedy algorithm.

The most dramatic difference is observed in the **Greedy Modularity** algorithm (Figure 10). In the unweighted network, the algorithm under-partitions the data into only 3 large, dispersed groups. However, when weights are introduced, it successfully identifies 6 distinct communities. This indicates that the Greedy algorithm is particularly sensitive to the "noise" of unweighted edges, requiring the weight signal to properly resolve the underlying social blocks.

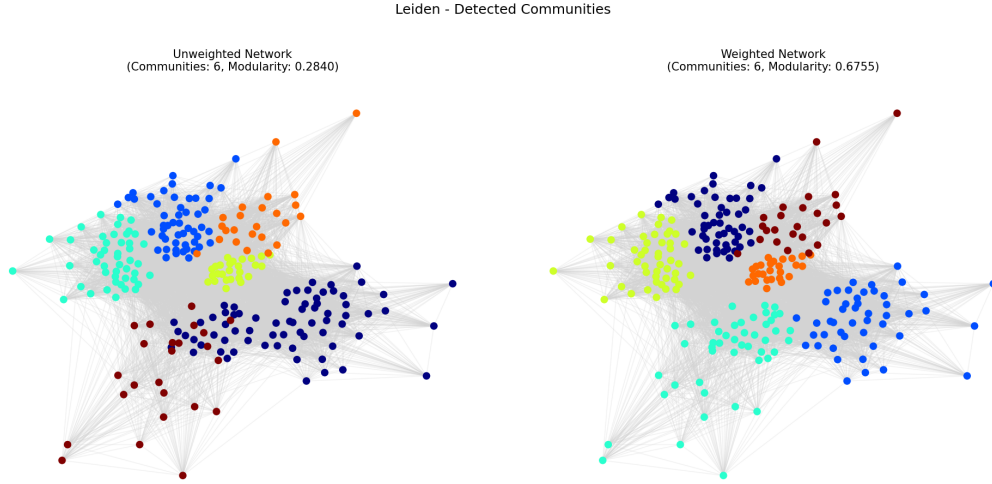


Figure 11: Community comparison for the Leiden algorithm.

4.2 Community Composition and Metadata

This section evaluates the relationship between the detected communities and the ground-truth metadata (10 classes and 1 teacher group). By comparing these partitions, we can determine how accurately the algorithms recover the school’s actual social organization.

Table 5: Validation against Ground Truth Metadata (11 School Groups).

Algorithm	Network	NMI	Jaccard	NVI
Louvain	Unweighted	0.8004	0.4553	0.1484
Louvain	Weighted	0.8051	0.4950	0.1466
Greedy	Unweighted	0.4308	0.1684	0.3345
Greedy	Weighted	0.8134	0.5016	0.1404
Leiden	Unweighted	0.8004	0.4553	0.1484
Leiden	Weighted	0.8051	0.4951	0.1466

As shown in Table 5, all algorithms achieve high Normalized Mutual Information (NMI) scores, particularly in the weighted network where values exceed 0.80. This indicates a strong alignment between the detected communities and the school classes. Interestingly, while Louvain and Leiden perform consistently across both network types, the **Greedy** algorithm shows a massive performance boost when weights are introduced, with its NMI nearly doubling from 0.43 to 0.81.

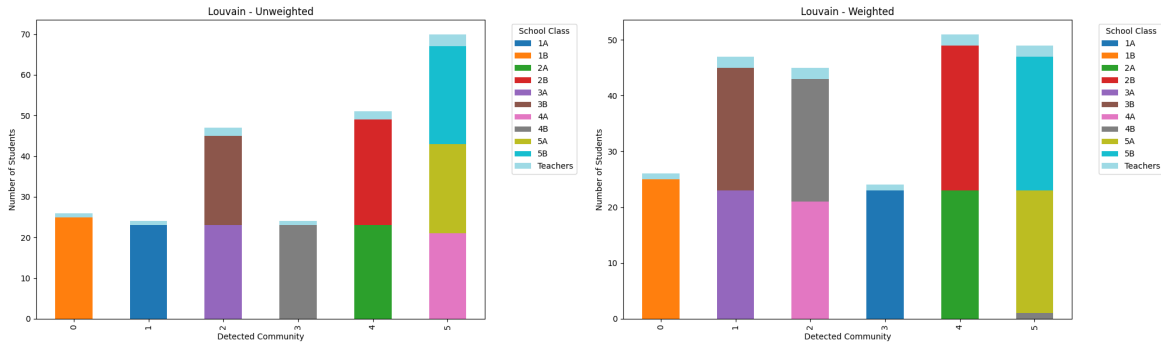


Figure 12: Community composition for Louvain: Unweighted (left) vs. Weighted (right).

The composition plots (Figures 12, 13, and 14) provide a visual explanation for these high scores. In the weighted networks, each community bar is dominated by one or two primary colors. This reveals that the algorithms are successfully identifying groups of students who interact frequently. The reason the NMI is not a perfect 1.0 is due to the resolution: while there are 11 ground-truth groups, the algorithms detect 6 communities, effectively merging the two classes of each grade (e.g., 1A and 1B) into a single social unit.

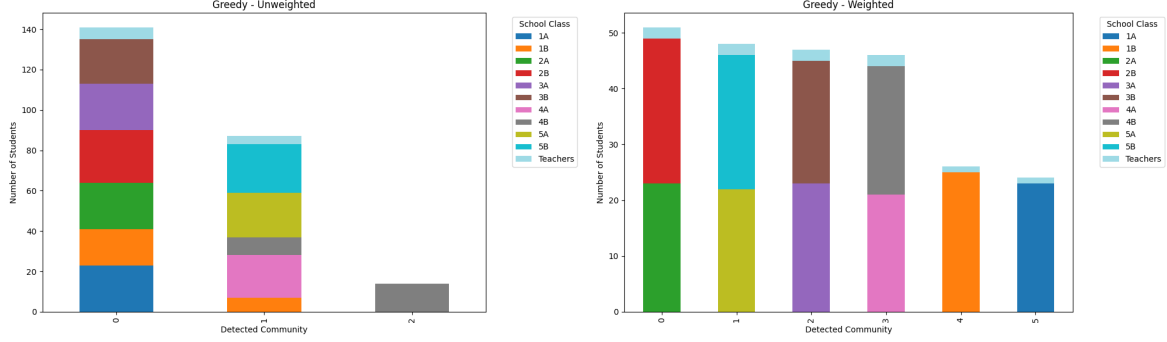


Figure 13: Community composition for Greedy: Unweighted (left) vs. Weighted (right).

The impact of interaction duration is most visible in the Greedy algorithm’s composition (Figure 13). In the unweighted version, the communities are highly mixed and ”noisy.” However, the weighted version produces communities with very high homogeneity, matching the performance of more complex algorithms like Louvain and Leiden.

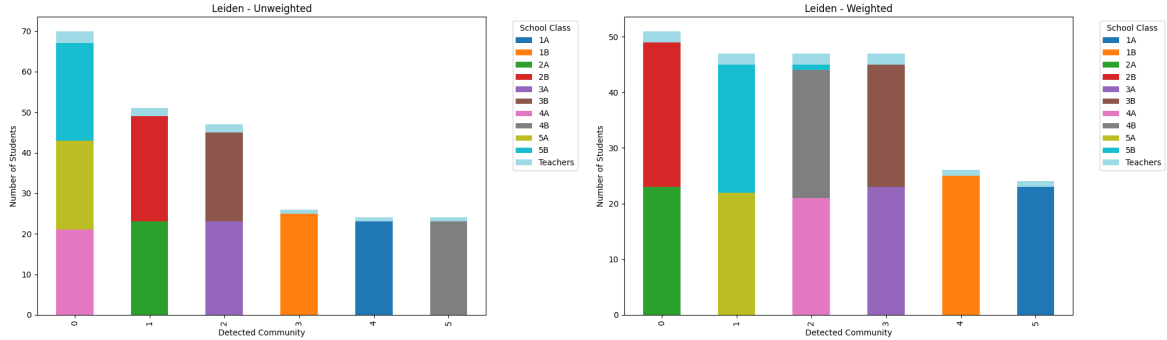


Figure 14: Community composition for Leiden: Unweighted (left) vs. Weighted (right).

In conclusion, face-to-face interaction time (weights) serves as a critical filter that reinforces class-based boundaries. While the algorithms tend to group students by grade level rather than individual class, the high NMI and low NVI values confirm that the detected structure is a highly accurate representation of the school’s actual social environment.

4.3 Discussion: The Relevance of Edge Weights

A key finding of this study is that including edge weights—which represent the total time students spent interacting—greatly improves how well the algorithms can find real communities.

Filtering Noise vs. Identifying Strength: In the unweighted version, a very short contact, such as students passing each other in a hallway, is treated the same as a long conversation in a classroom. This creates ”structural noise” that makes it harder to see the true social groups. As shown in Table 5, the NMI and Jaccard scores are lower when we ignore weights. Using weights acts as a filter, allowing the algorithms to focus on strong social bonds rather than accidental meetings.

The Sensitivity of the Greedy Algorithm: The most obvious proof of this is the Greedy Modularity algorithm. In the unweighted network, it had a low NMI of 0.4308 because it struggled to find the correct classes through the noise of many weak connections. However, when weighted data was used, its performance jumped to 0.8134, reaching the same level of accuracy as Louvain and Leiden. This result shows that simpler algorithms depend heavily on the quality of the data to find the best groups.

In summary, edge weights are essential for an accurate analysis of this social network. Without weights, the network appears as a messy "hairball" where boundaries are blurred. Using weighted data clarifies the social organization, which leads to a much stronger match between the detected groups and the actual school classes.

5 Conclusions

This study analyzed how community detection algorithms perform on synthetic and real-world networks. The results show that the accuracy of these methods depends heavily on the strength of the connections and the use of weights.

In the synthetic network tests, we found that algorithms only begin to identify the correct structure when the internal link probability (p_{rr}) is between 0.12 and 0.16. Below this point, the algorithms find many "fake" communities caused by random noise. A major finding is that a modularity value of $Q = 0.4$ can occur even in purely random networks. This proves that high modularity alone does not mean a network has a real community structure; it must be confirmed with external metrics.

In the primary school network, using edge weights (interaction time) was the most important factor for success. Unweighted networks include many accidental or brief contacts that confuse the algorithms. Using weights filters this noise and highlights real social bonds. This was most clear with the Greedy algorithm, where the NMI score jumped from 0.43 to 0.81 after adding weights. All algorithms successfully grouped students, mostly by their school grade level.

Finally, the choice of algorithm matters. Infomap is sensitive to small details but can over-partition noisy data, while Louvain and Leiden are more stable. To get reliable results in social networks, it is best to use weighted data and validate the findings with multiple accuracy metrics.